SANTIAGO RESTREPO RUIZ

Principal AI Engineer | Optimization, Cryptography & Electronics

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PROFILE

Computer scientist and electronic engineer, specialized in numerical control and representation learning, robotics, circuits, cryptography and reliable superintelligent systems.

EXPERIENCE

Principal System Architect (Artificial Intelligence)
PricewaterhouseCoopers (PwC)
May 2025 – July 2026 — Montevideo, Uruguay
- Leads AI system architecture for enterprise environments, aligning solution design with business strategy, security, and delivery constraints.
- Designs scalable data and application architectures for intelligent automation, analytics, and machine-learning workflows in regulated contexts.
- Translates complex technical and operational requirements into production-ready platforms, integration patterns, and implementation roadmaps.

Senior C++ Optimization Engineer
Globant
January 2023 – May 2025 — Remote, Colombia
- Stabilized and optimized a large-scale C/C++ platform for construction logistics, improving reliability and performance through applied mathematics and disciplined engineering practices.
- Led development of a high-performance real-time street-mapping and routing engine designed for global-scale deployment.

Senior Data Engineer (Banking & Retail)
Fourier Analytics
March 2021 – January 2023 — Remote, Colombia
- Designed and implemented data pipelines in C++ for high-volume banking and retail workloads.
- Built time-series forecasting and analytical models for operational and commercial decision-making.
- Secured APIs across teams while supporting compliance and data-protection requirements.

Lead Quantitative Analyst (Finance Engineering Lead)
AUB Capital
2020 – 2021 — Medellín, Colombia
- Led analytics and engineering work around a USD 10 million private-equity portfolio.
- Developed quantitative models and automated systems for low-risk global market instruments from concept through implementation.
- Built low-latency interfaces, secure Linux infrastructure, and forecasting workflows for financial engineering operations.

RPA Developer (Financial Process Automation)
Stratesys
March 2018 – June 2020 — Remote, Spain
- Delivered financial process automation (Automation Anywhere, UiPath, and Blue Prism).
- Developed supporting tooling and frameworks in Python, JavaScript, and PowerShell.
- Applied NLP and document-processing to multilingual classification and electronic-invoice extraction.

Founding Partner
Waajacu
January 2012 – Present — Colombia
- Founded and led an open-source organization focused on computational diplomacy, cryptography, optimization, and high-performance software engineering.
- Researches cryptographic protocols for cooperation, markets, law, and robotics.
- Explores entropy maximization, stochastic modeling, nonlinear dynamical systems, and optimal control.

EDUCATION

Electronic Engineering
Universidad del Quindío
February 2013 – September 2018 — Armenia, Colombia
Thesis: Design and Modeling of a Delta Robot.

Mathematics and Algorithms
Massachusetts Institute of Technology (OpenCourseWare)
February 2013 – September 2018 — Online
Independent study in algorithms, mathematics, and computer science through MIT OpenCourseWare.

SELECTED HIGHLIGHTS

- Top SABER PRO Engineering result (2018).
- Founder of www.waajacu.com: advanced engineering and research.
- Founder of Save the Bees and Seeds, an open-source software foundation.
- Thesis: Delta robot design and modeling.

CORE EXPERTISE

High-Performance C/C++
Robotics & Autonomous Systems
Embedded Systems & Electronics
Optimization & Optimal Control
Mathematical Modeling & Simulation
AI / Machine Learning Systems
LLM & Agentic AI Architecture
Applied Cryptography & Computational Diplomacy
Real-Time Systems & Algorithms
Data Engineering & Distributed Systems

TECHNICAL SKILLS

CORE STACK

Linux / Unix	●●●●●
Windows	●●●○○
C / C++	●●●●●
Python / JavaScript	●●●●●
Rust / Go / PHP	●●●○○
FPGA / Verilog / VHDL	●●●●○
SQL / NoSQL	●●●●○
Git | Docker / Kubernetes / Nginx	●●●●○
GDB / Valgrind	●●●○○
ADK / LangChain / LangGraph	●●●●○
PyTorch / TensorFlow	●●●●●

APPLIED DOMAINS

Mathematical Modeling	●●●●●
Entropy / Information Theory	●●●●○
Control / Geometry	●●●●○
Differential Privacy	●●●●○
Probability / Reinforcement Learning	●●●●○
Embedded Systems / Networking	●●●●○
Computational Diplomacy	●●●●○

LANGUAGES

Spanish	●●●●●
English	●●●●●
Chinese (Simplified)	●●●○○